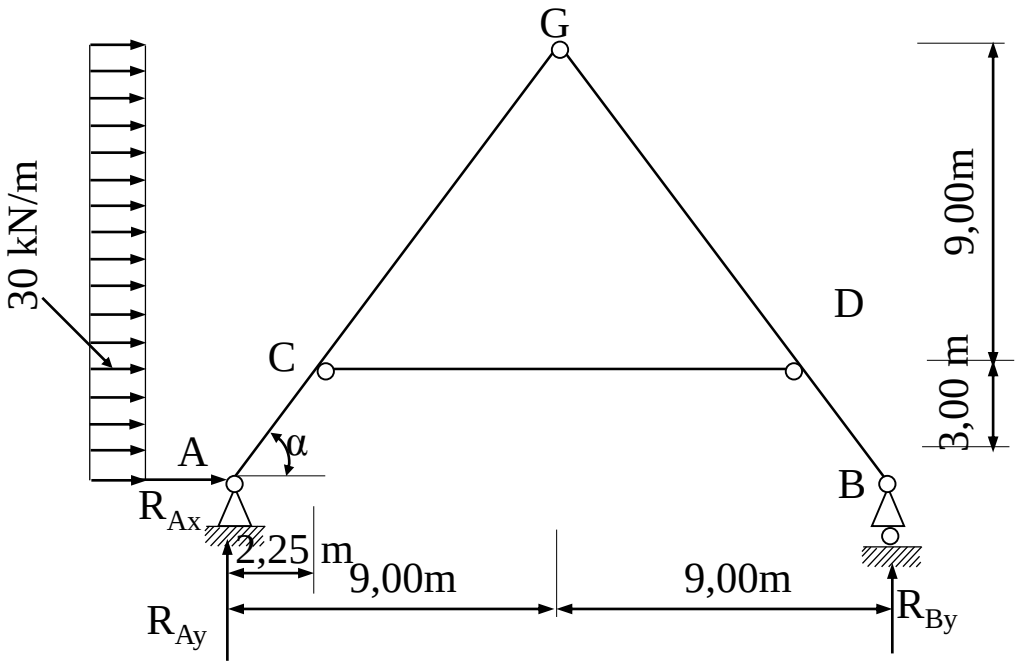
Frame with Tie-Rod

Example:



Estimate support reactions and draw internal force diagrams of the given system.

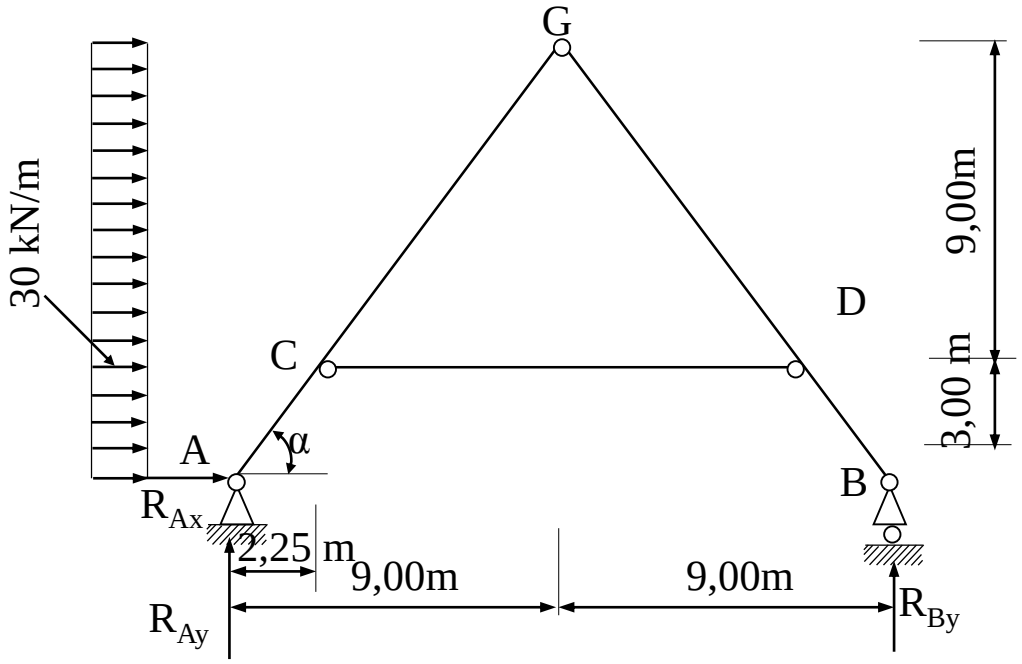
Solution:

Frame length and trigonometric identities

$$L_{AG} = L_{BG} = \sqrt{9^2 + 12^2} = 15{,}0 \text{ m}$$

$$\sin\alpha = 12/15 = 0{,}80 \qquad \cos\alpha = 9/15 = 0{,}60$$

$$L_{AC} = 3/0{,}8 = 3{,}75 \text{ m} \qquad L_{CG} = 9/0{,}8 = 11{,}25 \text{ m}$$



Support Reactions:

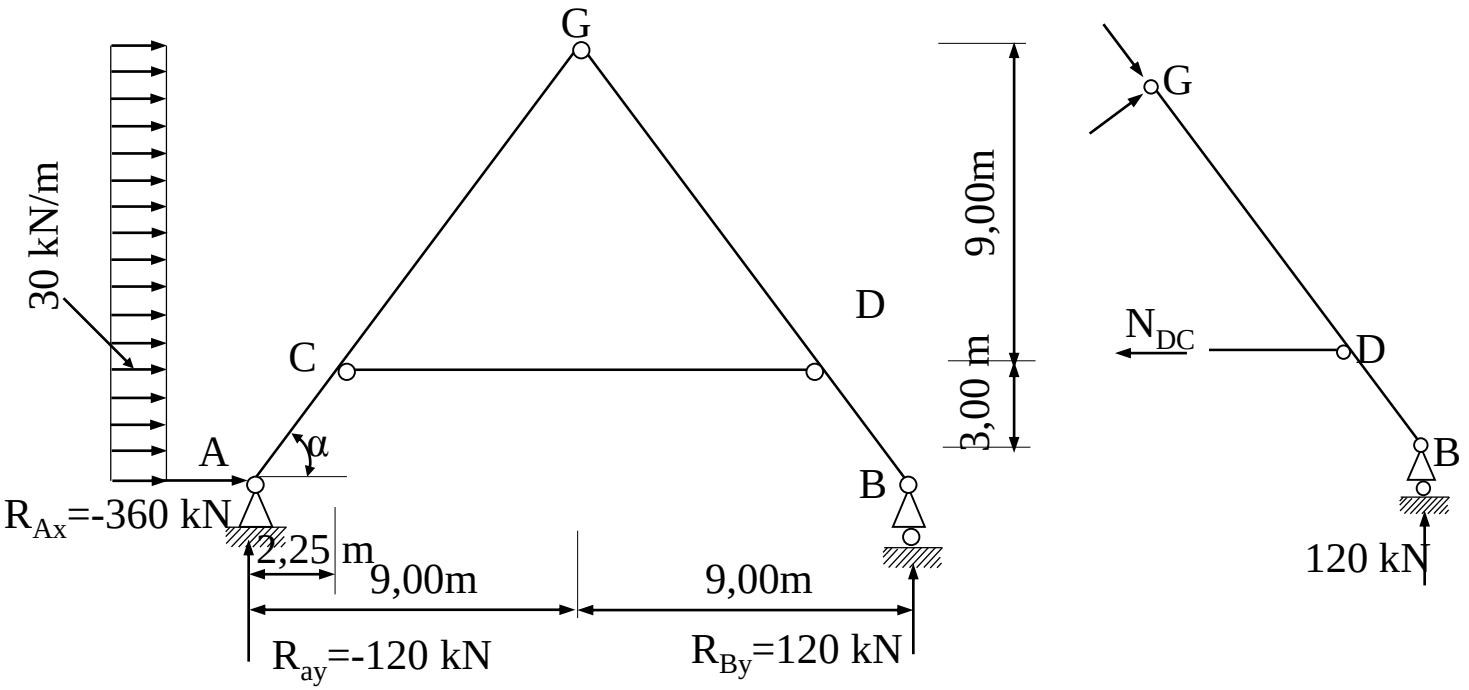
Whole system

$$\sum P_x = 0 \longrightarrow R_{Ax} = -30 \times 12{,}0 = -360 \text{ kN}$$

$$\sum M_B = 0 \longrightarrow R_{Ay} = -30 \times 12{,}0 \times 6{,}0 / 18{,}0 = -120 \text{ kN}$$

$$\sum M_A = 0 \longrightarrow R_{By} = 30 \times 12{,}0 \times 6{,}0 / 18{,}0 = 120 \text{ kN}$$

Check the system $\sum P_y = -120 + 120 = 0$



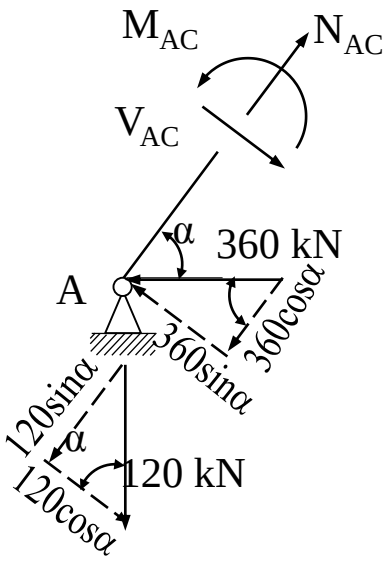
Tie-Rod Force

GDB system

$$\sum M_G = 0 \longrightarrow N_{DC} = 120 \times 9{,}0 / 9{,}0 = 120 \text{ kN}$$

$$N_{CD} = N_{DC} = 120 \text{ kN}$$

Internal Forces



A end of AC member

$$\sin\alpha = 0{,}80 \qquad \cos\alpha = 0{,}60$$

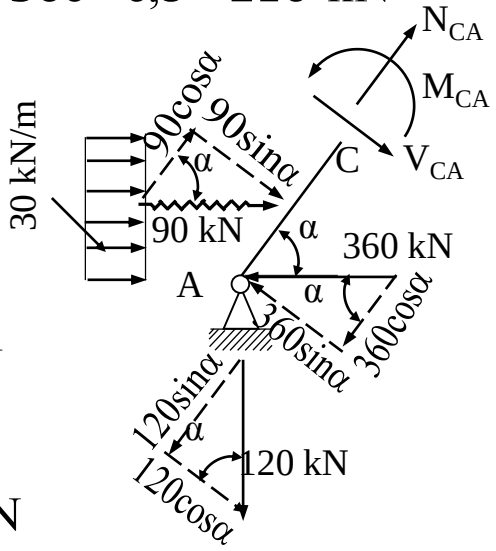
$$N_{AC} = 120\cdot\sin\alpha + 360\cdot\cos\alpha$$

$$N_{AC} = 120\times 0{,}8 + 360\times 0{,}6 = 312 \text{ kN}$$

$$V_{AC} = -120\cdot\cos\alpha + 360\cdot\sin\alpha$$

$$V_{AC} = -120\times 0{,}6 + 360\times 0{,}8 = 216 \text{ kN}$$

$$M_{AC} = 0$$



C end of AC member

$$N_{CA} = 120\cdot\sin\alpha + 360\cdot\cos\alpha - 90\cdot\cos\alpha$$

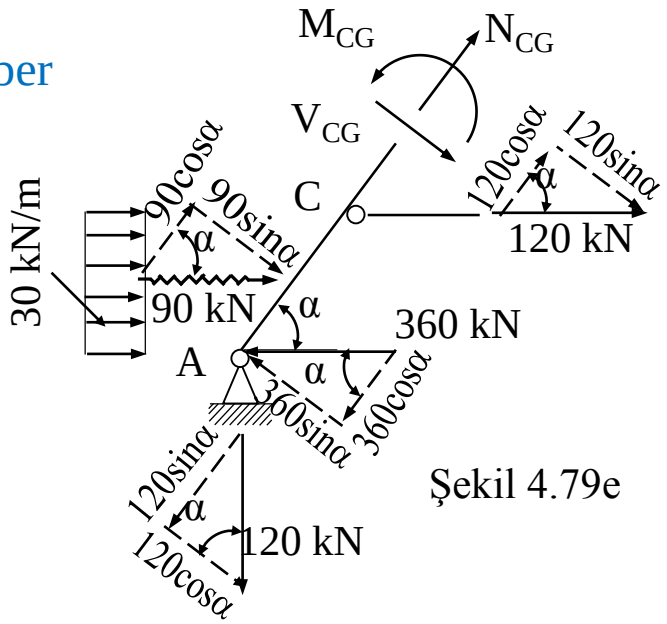
$$N_{CA} = 120\times 0{,}8 + (360-90)\times 0{,}6 = 258 \text{ kN}$$

$$V_{CA} = -120\cdot\cos\alpha + 360\cdot\sin\alpha - 90\cdot\sin\alpha$$

$$V_{CA} = -120\times 0{,}6 + (360-90)\times 0{,}8 = 144 \text{ kN}$$

$$M_{CA} = 360\times 3{,}0 - 120\times 2{,}25 - 90\times 1{,}5 = 675 \text{ kNm}$$

CG member



Şekil 4.79e

$$\sin\alpha = 0{,}80$$
$$\cos\alpha = 0{,}60$$

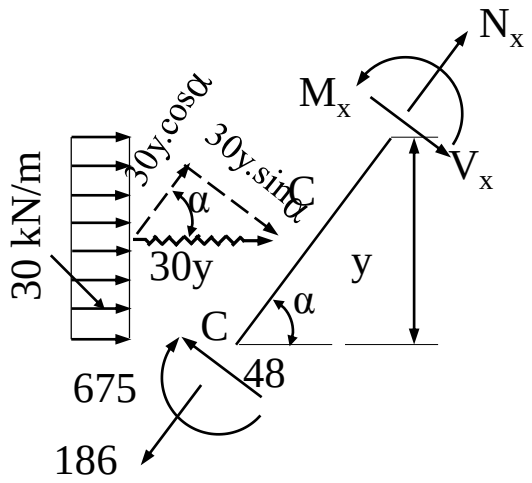
C end of CG member

$$N_{CG} = 120\cdot\sin\alpha + 360\cdot\cos\alpha - 90\cdot\cos\alpha - 120\cdot\cos\alpha$$
$$N_{CG} = 120\times 0{,}8 + (360-90-120)\times 0{,}6 = 186\ \text{kN}$$
$$V_{CG} = -120\cdot\cos\alpha + 360\cdot\sin\alpha - 90\cdot\sin\alpha - 120\cdot\sin\alpha$$
$$V_{CG} = -120\times 0{,}6 + (360-90-120)\times 0{,}8 = 48\ \text{kN} \qquad M_{CG} = M_{CA} = 675\ \text{kNm}$$

G end of CG member

$$N_{GC} = 120\cdot\sin\alpha + 360\cdot\cos\alpha - 360\cdot\cos\alpha - 120\cdot\cos\alpha$$
$$N_{GC} = 120\times 0{,}8 + (360-360-120)\times 0{,}6 = 24\ \text{kN}$$
$$V_{GC} = -120\cdot\cos\alpha + 360\cdot\sin\alpha - 360\cdot\sin\alpha - 120\cdot\sin\alpha$$
$$V_{GC} = -120\times 0{,}6 + (360-360-120)\times 0{,}8 = -168\ \text{kN} \qquad M_{GC} = 0$$

Since the sign of the shear forces at C and G ends of CG frame are different, shear force is zero at any location and moment is maximum at that point.



$$\sin\alpha = 0{,}80 \qquad \cos\alpha = 0{,}60$$

$$V_y = 48 - 30y.\sin\alpha = 48 - 30\times 0{,}8\cdot y =$$

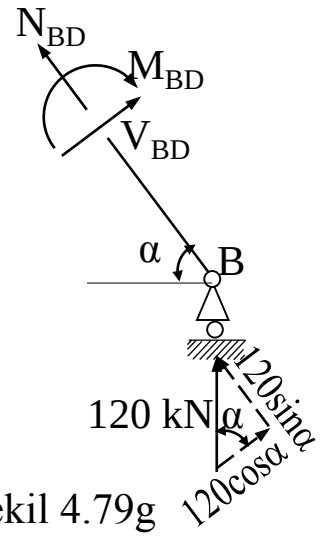
$$V_y = 48 - 24\cdot y = 0$$

$$y_0 = 48/24 = 2{,}0 \ \text{ m}$$

$$M_y = 675 + 48\times(y/\sin\alpha) - 30\cdot y\cdot y/2 = 675 + 60y - 15y^2$$

$$M_{enb.} = 675 + 60\times 2{,}0 - 15\times 2^2 = 735 \ \text{kNm}$$

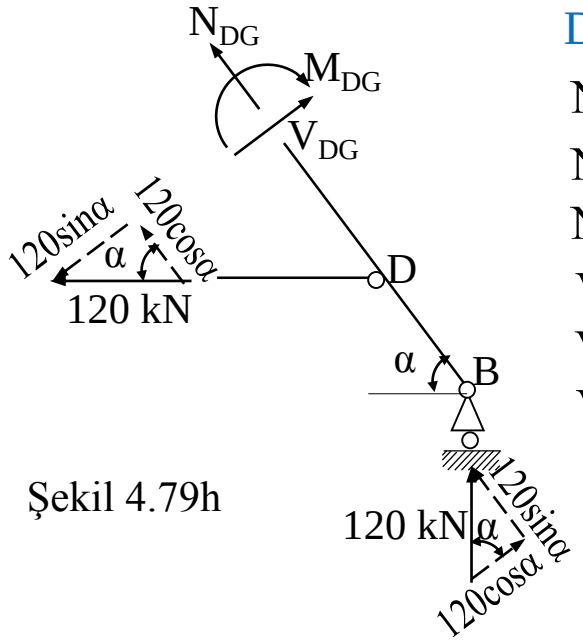
$\sin\alpha = 0{,}80 \qquad \cos\alpha = 0{,}60$



Şekil 4.79g

DB member

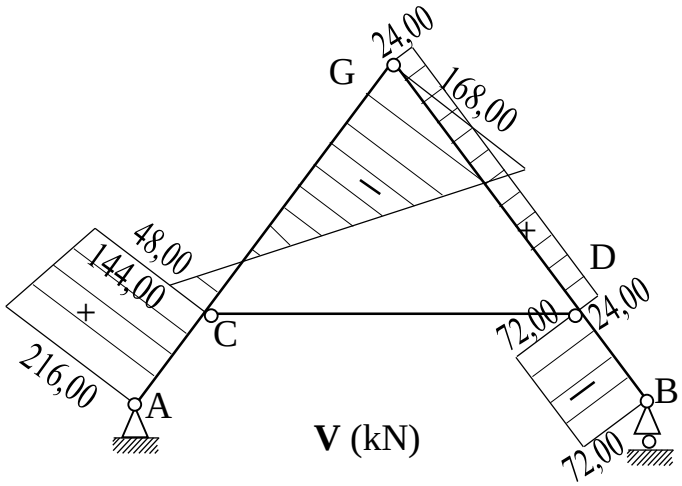
$$N_{BD} = N_{DB} = -120 \cdot \sin\alpha$$
$$N_{BD} = N_{DB} = -120 \times 0{,}8 = -96 \ \text{kN}$$
$$V_{BD} = V_{DB} = -120 \cdot \cos\alpha$$
$$V_{BD} = V_{DB} = -120 \times 0{,}6 = -72 \ \text{kN}$$
$$M_{BG} = 0 \qquad M_{DB} = 120 \times 2{,}25 = 270 \ \text{kNm}$$



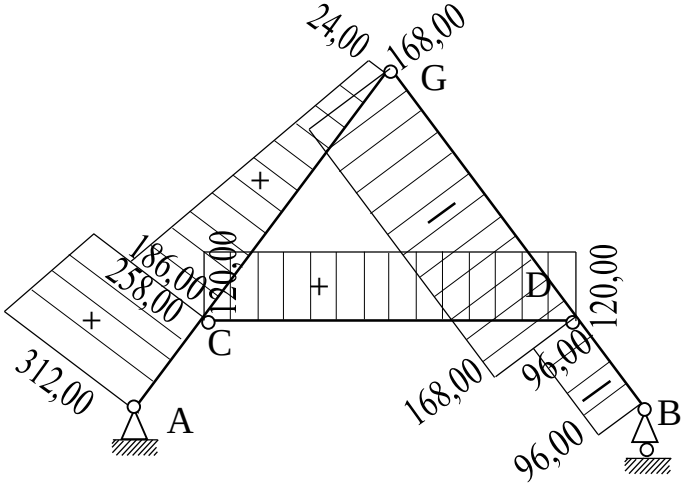
Şekil 4.79h

DG member

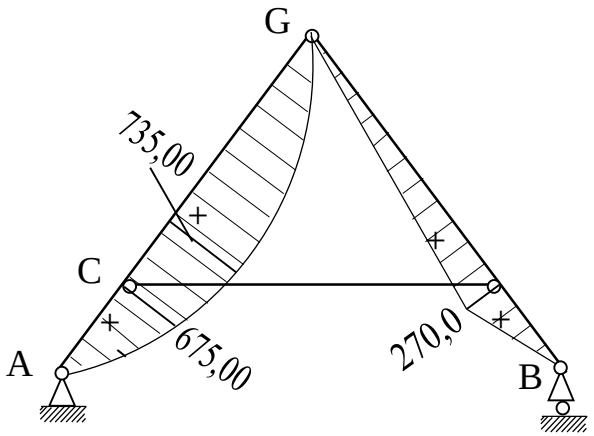
$$N_{DG} = N_{GD} = -120 \cdot \sin\alpha - 120.\cos\alpha$$
$$N_{DG} = N_{GD} = -120 \times 0{,}8 - 120 \times 0{,}6$$
$$N_{DG} = N_{GD} = -168 \ \text{kN}$$
$$V_{DG} = V_{GD} = -120 \cdot \cos\alpha + 120\sin\alpha$$
$$V_{DG} = V_{Gd} = -120 \times 0{,}6 + 120 \times 0{,}8$$
$$V_{DG} = V_{Gd} = 24 \ \text{kN}$$
$$M_{DG} = M_{DB} = 270 \ \text{kNm}$$
$$M_{GB} = 0$$


G
24,00
168,00
48,00
144,00
216,00
C
D
72,00
24,00
A
B
72,00

V (kN)


24,00
168,00
G
186,00
258,00
120,00
120,00
312,00
C
D
168,00
96,00
96,00
A
B

N (kN)


G
735,00
C
A
675,00
270,0
B

M (kNm)

Reference:

- Prof. İbrahim EKİZ, YAPI STATİĞİ 1, BİRSEN YAYINEVİ

Acknowledgement:

Thank you to Prof. İbrahim EKİZ for sharing this presentation with me.